4 A receiver aerial is placed between a point source emitting microwaves at a single fixed wavelength of 140 mm and a reflective metal sheet, as shown in Figure 4.1.

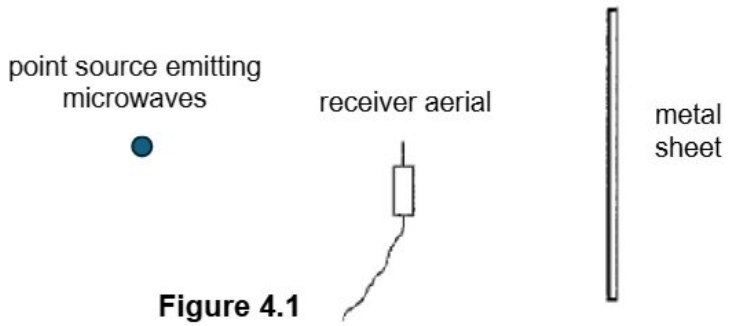


Figure 4.1

The receiver aerial is fixed in position. The metal sheet is moved away from the receiver aerial toward the right. The detected voltage at the receiver aerial varies with the distance moved by the sheet as shown in Figure 4.2.

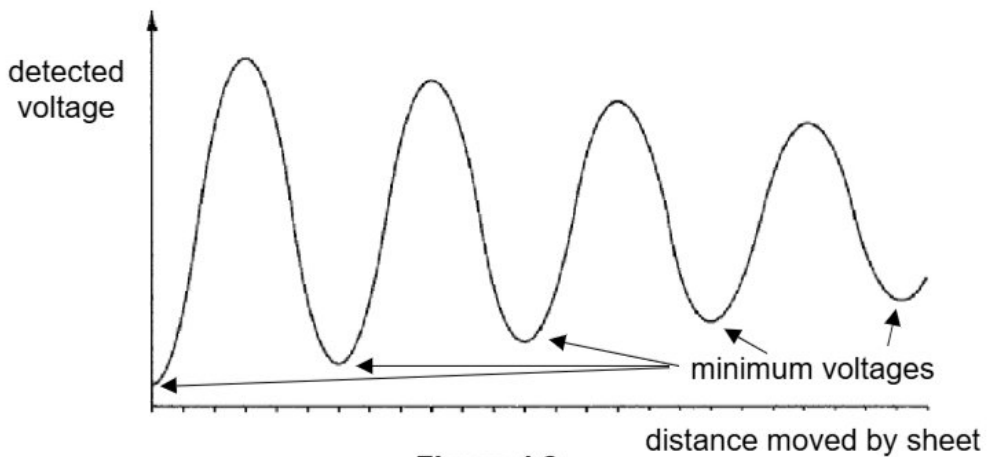


Figure 4.2

(a) Explain why the minimum voltage detected at the receiver aerial gets larger with distance moved by the sheet.

.....

.....

.....

.....

.....

.....

.....

..... [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(b) State the distance moved by the metal plate between consecutive detection by the aerial receiver of minimum voltages.

distance = ... mm [1]

[Total: 4]